# UNITED INTERNATIONAL UNIVERSITY (UIU)

Dept. of Computer Science & Engineering

Class Test 1 (Set E)

Course No: CSE 4325 Course Title: Microprocessors and Microcontrollers

**Time: 15 minutes** **Marks: 20**

| **Name** | | **Section** | | **ID** | |
|---|---|---|---|---|---|

1. Transfer of bus control from the processor to the device takes **450 ns**, while transfer of bus control from the device to the processor takes **350 ns**. If one of the I/O devices uses DMA in **cycle stealing mode** and has a data transfer rate of **35 KB/s**, how long will it take to transfer a block of **540 bytes**? Assume that **6 bytes** are transferred at a time in this mode. (10)

   **Ans.** $(450 + (6 \times 10^9) / (35 \times 2^{10}) + 350) \times (540 / 6) = 15138964.29$ ns

2. Transfer of bus control from the processor to the device takes **600 ns**, while transfer of bus control from the device to the processor takes **500 ns**. One of the I/O devices uses DMA **in both burst mode and cycle stealing mode** and has a data transfer rate of **50 KB/s** For the **first half** of the bytes, the device uses burst mode and for the rest half, it uses cycle stealing mode. Assume that in cycle stealing mode, data is transferred **16 bytes** at a time. How many bytes are transferred in **80141900 ns**? (10)

   **Ans.**
   let, Total data = **X bytes**, So, **X/2 bytes** are sent in burst mode and **X/2 bytes** are sent in cycle stealing mode.

   $(600 + ((\mathbf{X}/2) \times 10^9) / (50 \times 2^{10}) + 500)$
   $+ (600 + (16 \times 10^9) / (50 \times 2^{10}) + 500) \times (\mathbf{X}/2) / 16 = 80141900$
   $\Rightarrow 1100 + 9765.625\mathbf{X} + 9800\mathbf{X} = 80141900$
   $\Rightarrow 19565.625\mathbf{X} = 80140800$
   $\Rightarrow X = 4096$ bytes.

**UNITED INTERNATIONAL UNIVERSITY (UIU)**
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Class Test 1 (Set F)
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**Time: 15 minutes** **Marks: 20**

| **Name** | | **Section** | | **ID** | |
|---|---|---|---|---|---|

1. Transfer of bus control from the processor to the device takes **600 ns**, while transfer of bus control from the device to the processor takes **500 ns**. One of the I/O devices uses DMA **in both burst mode and cycle stealing mode** and has a data transfer rate of **50 KB/s** For the **first half** of the bytes, the device uses burst mode and for the rest half, it uses cycle stealing mode. Assume that in cycle stealing mode, data is transferred **16 bytes** at a time. How many bytes are transferred in **80141900 ns**? (10)

   **Ans.**
   let, Total data = **X bytes**, So, **X/2 bytes** are sent in burst mode and **X/2 bytes** are sent in cycle stealing mode.

   $(600 + ((\mathbf{X}/2) \times 10^9) / (50 \times 2^{10}) + 500)$
   $+ (600 + (16 \times 10^9) / (50 \times 2^{10}) + 500) \times (\mathbf{X}/2) / 16 = 80141900$
   $\Rightarrow 1100 + 9765.625\mathbf{X} + 9800\mathbf{X} = 80141900$
   $\Rightarrow 19565.625\mathbf{X} = 80140800$
   $\Rightarrow X = 4096$ bytes.

2. Transfer of bus control from the processor to the device takes **750 ns**, while transfer of bus control from the device to the processor takes **650 ns**. If one of the I/O devices uses DMA in **cycle stealing mode** and has a data transfer rate of **35 KB/s**, how long will it take to transfer a block of **544 bytes**? Assume that **8 bytes** are transferred at a time in this mode. (10)

   **Ans.** $(750 + (8 \times 10^9) / (35 \times 2^{10}) + 650) \times (544 / 8) = 15273771.43$ ns